Mean Price in a Lot-Based Inventory Simulation: A Comparison of Issuing Strategies

Thomas Kausch

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Abstract

We describe a discrete-event simulation of a single-item inventory with (s, S) replenishment in which stock is held in lots with individual unit prices. The rule that decides which lot an item is taken from, the *issuing strategy*, is a pluggable component. We compare four strategies: cheapest-first, FIFO, LIFO, and an alternation between the cheapest and the most expensive lot. The mean price of the items in stock is tracked after every event. Under cheapest-first issuing it is a biased and therefore unusable indicator of purchase prices: expensive lots are served last, accumulate in the stock, and push the mean price towards the upper end of the lot-price distribution. The other three strategies keep it close to the mean lot price.

1 Model

The simulation is implemented in Python with SimPy and consists of three components.

Stock. The stock is a set of lots. Each lot j has a quantity q_j, a single unit price p_j shared by all of its items, and an arrival number a_j. The initial stock is one lot whose price equals the expected lot price μ_p.

Demand. Demand orders arrive as a Poisson process with rate λ, i.e. with exponentially distributed inter-arrival times. The size of an order is $\lceil X \rceil$ with $X \sim \mathcal{N}(\mu, \sigma^2)$, bounded below by 1. Each item of an order is taken from a lot chosen by the issuing strategy (Section 3). Demand that cannot be served is counted as lost.

Replenishment. We use an (s, S) policy with $s = S\,(1 - \tau)$, where τ is the fraction by which the stock may be empty before a re-order is triggered. After every demand order, the inventory position (on hand plus on order) is checked. If it is at or below s, an order of size $S -$ position is placed. It arrives after an exponentially distributed lead time as *one* lot whose unit price, the *market price* at the time of purchase, is drawn from $\mathcal{N}(\mu_p, \sigma_p^2)$ (truncated at 0.01).

2 Computation of the Mean Price

Let $L(t)$ be the set of lots in stock at time t. The number of items and their total value are

$$N(t) = \sum_{j \in L(t)} q_j, \qquad V(t) = \sum_{j \in L(t)} q_j\, p_j, \tag{1}$$

and the mean price per item in stock is

$$\bar{p}(t) = \frac{V(t)}{N(t)}. \tag{2}$$

Symbol	Meaning	Value
T	simulated time	800
λ	demand arrival rate	2.0
$\mu,\ \sigma$	order size mean / std. dev.	5.0 / 2.0
S	order-up-to level (capacity)	1600
τ	re-order trigger fraction	0.25
s	re-order point $S(1-\tau)$	1200
	mean lead time	5.0
	initial stock / initial price	1400 / μ_p
$\mu_p,\ \sigma_p$	lot price mean / std. dev.	10.0 / 5.0
	random seed	42

Table 1: Parameters of the simulation runs shown in this paper.

The sums are maintained incrementally: when a lot of quantity q and price p arrives, $N \leftarrow N+q$ and $V \leftarrow V + q\,p$; when one item is taken from lot j, $N \leftarrow N-1$ and $V \leftarrow V - p_j$, and an exhausted lot is removed. After every change the triple $(t, N, \bar{p})$ is appended to a history, so $\bar{p}$ is a piecewise-constant function of time. When the stock is empty, V is reset to zero to remove accumulated floating-point error and $\bar{p}$ is undefined.

3 Issuing Strategies

An issuing strategy is a function that, given the lots in stock, returns the lot from which the *next single item* is taken. Choosing per item rather than per order makes it possible to split an order across lots according to a rule. All strategies serve the same demand from the same purchases; only the choice of lot differs.

Cheapest-first. Take from the lot with the lowest price, $\arg\min_j p_j$ (ties: earlier arrival). Expensive lots are touched only when all cheaper ones are gone.

FIFO. Take from the oldest lot, $\arg\min_j a_j$. The price plays no role.

LIFO. Take from the newest lot, $\arg\max_j a_j$. The oldest lots stay at the bottom of the stack.

Cheapest/priciest. Alternate item by item between the cheapest lot $\arg\min_j p_j$ and the most expensive lot $\arg\max_j p_j$, so that both ends of the price range are drawn down by the same amount. This strategy is stateful (it remembers which end is next).

4 Results

All runs use the parameters of Table 1, the same seed, and therefore the same 8 719 demanded items and the same 21 purchases (mean realised lot price 11.5, maximum 21.7). No demand is lost. Table 2 summarises the mean price of the stock, and Figure 1 shows its course for all four strategies together with the market prices of the purchases.

Cheapest-first. Figure 2 shows this run in detail. Lot prices are drawn around $\mu_p = 10$, yet $\bar{p}$ climbs to about 20, twice the expected purchase price and close to the maximum lot price. This is a structural consequence of the rule. A lot is touched only after all cheaper lots are exhausted, and under an (s, S) policy the stock is refilled long before it is empty (the stock never fell below 973 of 1600 items). Cheap lots are therefore consumed right after delivery, while expensive lots stay and form a sediment that dominates $\bar{p}$. At the end of the run the stock consists of four

Strategy	End	Average	Min	Max
Cheapest-first	19.53	15.64	10.00	20.14
FIFO	11.22	11.37	7.27	15.11
LIFO	10.66	10.56	8.41	13.82
Cheapest/priciest	10.95	11.09	9.33	13.87

Table 2: Mean price $\bar{p}$ of the stock at the end of the run and its average, minimum and maximum over all events.

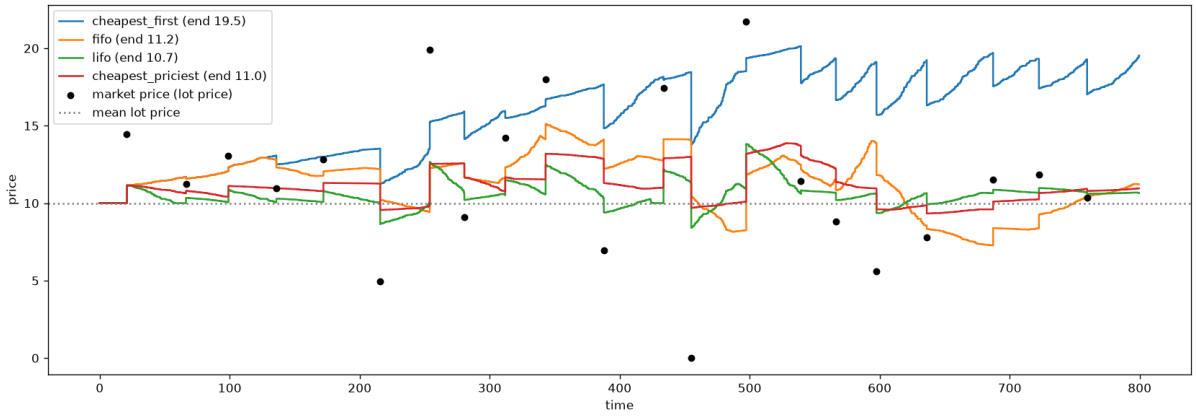


Figure 1: Mean price of the stock under the four issuing strategies. The black dots are the market prices (lot prices) of the purchases, the dotted line is the expected lot price $\mu_p = 10$.

lots, priced 11.85, 17.98, 19.88 and 21.71, the last one being the most expensive purchase of the whole run, delivered at $t \approx 497$ and still untouched.

FIFO. The oldest lot is always served first, so a lot stays in stock for about one replenishment cycle, regardless of its price. $\bar{p}$ follows the recent purchase prices with a delay and fluctuates between 7.3 and 15.1 around the expected lot price, without a trend. It ends at 11.22, close to the realised mean lot price of 11.5.

LIFO. The newest lot is served first, so the stock is split into a small, quickly turning top and an untouched bottom. Because the initial lot was priced at μ_p, the untouched bottom (973 items at 10.00 at the end) happens to sit exactly at the expected price, and $\bar{p}$ has the smallest spread of all strategies (8.4 to 13.8, end 10.66). This is an artefact of the initial price: with an initial lot at 15 instead of 10, the same run ends at 14.9. LIFO thus freezes whatever lots happen to be at the bottom, here the initial lot.

Cheapest/priciest. Drawing equal amounts from both ends removes the sediment at the expensive end as well as the cheap lots. $\bar{p}$ stays between 9.3 and 13.9 and ends at 10.95, the narrowest range together with LIFO, and, unlike LIFO, without depending on the initial lot, because both extremes are consumed.

5 Consequences

Under cheapest-first issuing the mean price of the stock is a *survivorship-biased* quantity: it describes the items that nobody wanted to take, not the items that were bought.

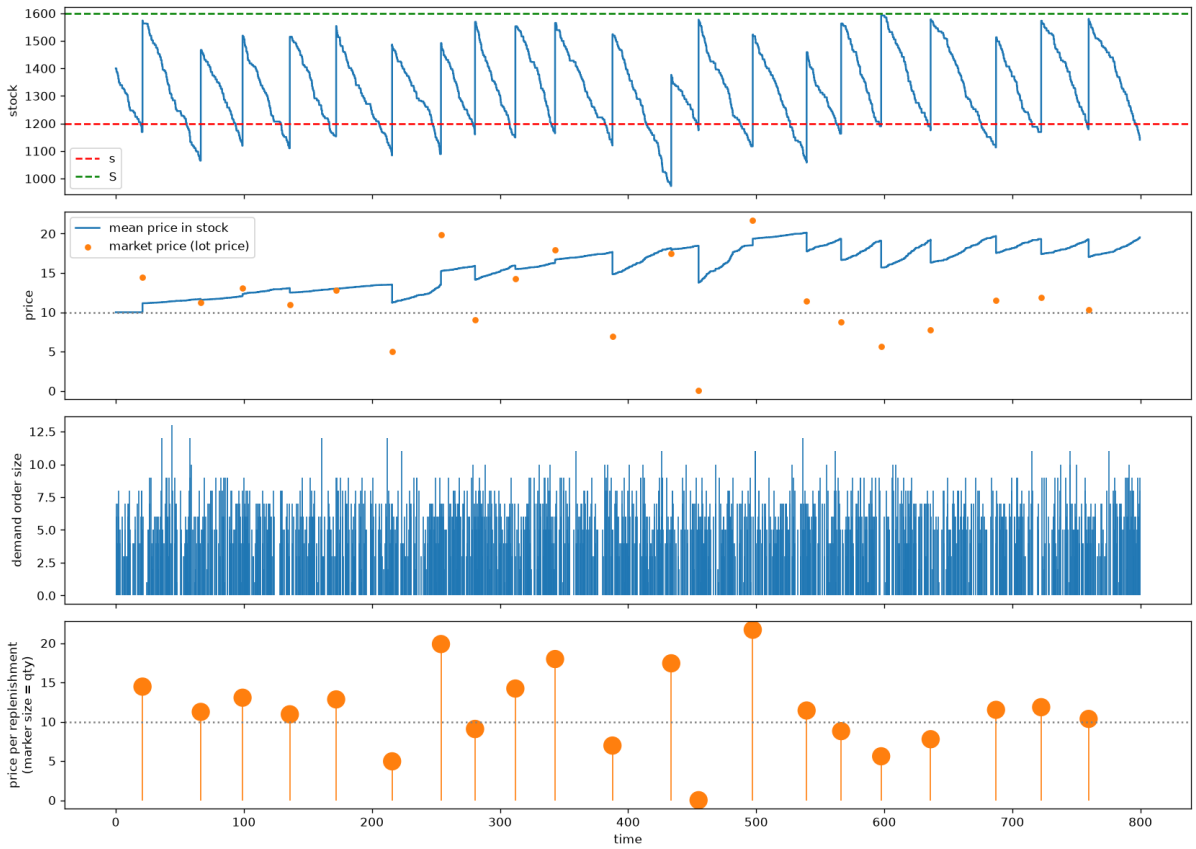


Figure 2: Cheapest-first issuing. From top to bottom: stock level with re-order point s and capacity S; mean price (line) and market prices of the purchases (dots); demand order sizes; lot prices, marker area proportional to lot size.

- It is **not an estimator of the purchase price**: its long-run value depends on the number of purchases and the spread σ_p, not on μ_p.
- It is **not usable as a valuation or cost basis** for the issued items, since these are by construction the cheapest ones.
- It is **not a stable state variable**: it trends with simulation length and depends on single expensive lots.
- It carries **no information about the current market price** and cannot drive re-ordering or pricing decisions.

FIFO, LIFO and cheapest/priciest do not have this drift, because they do not let one price class accumulate in the stock. They are not equally robust, though. FIFO and cheapest/priciest depend only on the recent purchases, while LIFO depends on the lots left at the bottom of the stack, here the initial lot.

Limits. A balanced mean price does not guarantee a profit. Items taken from expensive lots, as under cheapest/priciest, are sold with a low margin or at a loss if the selling price does not follow the purchase price. Fixing the bias of the statistic therefore opens a second question, the profitability of the rule.

Lesson learned. In reality the mean price influences the selling price and with it the demand. Here demand is independent of the price: λ and the order size are constants. A realistic model

must close this loop and describe demand as a function of the (mean) price, for example $\lambda = \lambda(\bar{p})$. Only then can the issuing strategies be compared by their effect on sales volume and profit and not only by the mean price they produce.

6 Conclusion

We presented a lot-based inventory simulation with (s, S) replenishment and a pluggable issuing strategy, and the incrementally maintained mean price $\bar{p} = V/N$. Cheapest-first issuing gives this measure a systematic upward bias: expensive lots are never consumed, remain in stock, and drive $\bar{p}$ towards the maximum lot price (19.5 against an expected 10). FIFO, LIFO and the alternation between the cheapest and the most expensive lot keep it between about 7 and 15 around the expected lot price. The problem is therefore not the mean price itself but the decision to always serve from the cheapest lot, and the resulting property cannot be used as a price indicator. A comparison of the strategies by profit requires price-dependent demand and a selling price, which the present model does not have.